

Ecuaciones en Recurrencia

Caso A.

$$T(n) = aT(n-b) + n^d \log^p n$$

$$\Theta(T(n)) = \begin{cases} a^{n/b} \log^p n, & \text{si } a > 1 \\ n^{d+1} \log^p n, & \text{si } a = 1 \\ n^d \log^p n, & \text{si } a < 1 \end{cases}$$

Caso B.

$$T(n) = aT(n/b) + n^d \log^p n$$

$$\Theta(T(n)) = \begin{cases} n^{\log_b a}, & \text{si } a > b^d \\ n^d \log^{p+1} n, & \text{si } a = b^d \\ n^d \log^p n, & \text{si } a < b^d \end{cases}$$

Aproximación.

$$T(n) = a_1 T(n-b_1) + \dots + a_k T(n-b_k) + c_1^n g_1(n) + \dots + c_j^n g_j(n)$$

siendo $b_i < b_{i+1}$ y $c_i < c_{i+1}$

$$T(n) <_{\infty} (a_1 + \dots + a_k) T(n-b_1) + c_j^n g(n)$$

$$T(n) >_{\infty} (a_1 + \dots + a_k) T(n-b_k) + c_j^n g(n)$$

Aproximación.

$$T(n) = a_1 T\left(\frac{n}{b_1}\right) + \dots + a_k T\left(\frac{n}{b_k}\right) + g(n)$$

siendo $b_i < b_{i+1}$

$$T(n) <_{\infty} (a_1 + \dots + a_k) T\left(\frac{n}{b_1}\right) + g(n)$$

$$T(n) >_{\infty} (a_1 + \dots + a_k) T\left(\frac{n}{b_k}\right) + g(n)$$

Propiedades de los logaritmos

$$a^{\log_a n} = n$$

$$\log_a n = \frac{1}{\log_b a} \log_b n$$

$$\log_a n = x \equiv a^x = n$$

$$\log_a n > 0 \text{ si } n > 1$$

Propiedades de los sumatorios

$$\sum_{i=a}^b f_i = f_a + f_{a+1} + \dots + f_b$$

$$\sum_{i=a}^b (f_i + g_i) = \sum_{i=a}^b f_i + \sum_{i=a}^b g_i$$

$$\sum_{i=a}^b \sum_{j=c}^d f(i)g(j) = \sum_{i=a}^b f(i) \sum_{j=c}^d g(j)$$

$$\sum_{i=a}^b g(x)f(i) = g(x) \sum_{i=a}^b f(i), \text{ si } g(x) \text{ no depende de } i$$

$$\sum_{i=a}^b f(i) = \sum_{i=a}^c f(i) - \sum_{i=a}^{b-1} f(i), \text{ con } c > b$$

Algunos sumatorios usuales

$$\sum_{i=a}^n 1 \cong n$$

$$\sum_{i=a}^n i = \frac{n(n+1)}{2} \approx \frac{1}{2}n^2 + \dots$$

$$\sum_{i=a}^n i^2 = \frac{n(n+1)(2n+1)}{6} \approx \frac{1}{3}n^3 + \dots$$

$$\sum_{i=a}^n i^k \approx \frac{1}{k+1} n^{k+1} + \dots$$

$$\left. \begin{array}{l} |\{a, a+r, a+2r, \dots, n\}| \\ |\{n, n-r, n-2r, \dots, a\}| \end{array} \right\} \sum_{x \in pa(a,r)}^n x^d \log^p x$$

$$\left. \begin{array}{l} |\{a, ar, ar^2, \dots, n\}| \\ |\{n, n/r, n/r^2, \dots, a\}| \end{array} \right\} \sum_{x \in pg(a,r)}^n x^d \log^p x$$

$$\sum_{x \in pa(a,r)}^n x^d \log^p x =_{\infty} \frac{1}{r(d+1)} n^{d+1} \log^p n$$

$$\sum_{x \in pg(a,r)}^n x^d \log^p x \cong_{\infty} \begin{cases} \log^{p+1} n, & \text{si } d = 0, r > 1 \\ n^d \log^p n, & \text{si } d > 0, r > 1 \end{cases}$$